Exercise 01: Introduction

Task 1*: Schema architecture

You are tasked with developing a management system for the examination grades at the TH Rosenheim. There are only two types of users of this system: students, who are of course only allowed to see their own grades, and professors, who enter the grades for the lectures they give.

a) What could the two *external* schemas look like for these two user groups? Draw possible tables as a solution. Remember: an external schema is the output, that a database generates for a specific use case or app. **(5 Points)**
b) What could a *conceptual* schema look like? Draw possible tables as a solution. Remember, a conceptual schema is how we, as the database developers, structure the data in tables or relations. **(5 Points)**
c) What could an *internal* schema look like? Answer the question in one sentence. Remember: an internal schema is how the database internally stores the data.

Task 2*: Set up your environment

For the further participation on this course, it is very important to have a running development environment. Please follow the instructions in the Learning Campus, to do a setup. Please approach us for further options. Since it is based on docker, the setup will be available for various operating systems. Run the provided sql script to test your environment.

Use ChatGPT to generate a query that provides the average grades by providing the given sql script that contains the table definitions to ChatGPT by using the prompt: “create a query for the given file in tsql that provides the average grades for all students.” Extend the prompt so that the rounded grades are generated, since there is no grade 1.15, but only the grades 1.0, 1.3, 1.7, 2.0, 2.3, 2.7, 3.0, 3.3, 3.7, 4.0 and 5.0.

Task 3: Problems with data redundancy

You live in a shared apartment with two other students, Julia and Christian. Each person has their own room, and in addition you have a shared living room and kitchen. Because you get on well with each other, you do a lot of things together, cook together, celebrate together and also cook for each other. So there's a lot going on in your shared apartment!

a. List three very specific problems that this scheduling process can cause for the three of you.
b. Which solution would you suggest?

Task 4*: Data independence

Your management system from Task 3 is very successful, so the Examinations Office contacts you and would also like to be able to access your database as a third user group. The Examinations Office needs access to all data, so you create a third suitable external schema for this purpose.

a) Explain what *logical data independence* means here. **(2 Points)**
b) Due to the numerous large accesses by the Examinations Office, your database becomes very slow. So you decide to optimise the internal representation. Explain what *physical data independence* means here. What significant advantage does this have? **(2 Points)**